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Modular sensor platform apparatus

Abstract

Provided is a modular sensor platform apparatus. The modular sensor platform apparatus includes at least one first terminal capable of being coupled to a power sensor, a plurality of second terminals capable of being coupled to an external sensor module and a communication module, and a micro-control unit (MCU) configured to execute a predefined algorithm.

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Background/Summary

TECHNICAL FIELD
(1) An embodiment of the present disclosure relates to a modular sensor platform apparatus, and more particularly, to an Internet of Things (IoT)-based modular sensor platform apparatus for use in a smart engineering solution.
BACKGROUND ART
(2) A conventional machine tool power measurement device includes a simple circuit that collects power data and transmits the same through a wire. In addition, conventional devices have insufficient data per unit time to analyze the quality and conditions of machine tools. Conventional devices are intended to measure power rather than to analyze various characteristics of machine tools. Such conventional devices do not conform to the direction of development of smart engineering devices capable of measuring various physical quantities at the same time and calculating various physical quantities of machine tools at a time.
(3) Moreover, when a conventional power measurement apparatus and various sensor systems are attached to one machine tool for measurement, a problem of data synchronization occurs and a measurement value of each sensor needs to be transmitted wirelessly, which increases a traffic volume. In addition, as computation becomes impossible at an end of a system, the efficiency of edge computing as one of the core techniques of smart engineering may not be achieved.
DISCLOSURE
Technical Problem
(4) An embodiment of the present disclosure provides a modular sensor platform apparatus easily applicable to a smart engineering solution by simultaneously serving as an IoT device and for edge computing.
Technical Solution

(5) According to an embodiment of the present disclosure, a modular sensor platform apparatus includes at least one first terminal capable of being coupled to a power sensor, a plurality of second terminals capable of being coupled to an external sensor module and a communication module, and a micro-control unit (MCU) configured to execute a predefined algorithm.

Advantageous Effects

(6) According to an embodiment of the present disclosure, a modular sensor platform apparatus may use a non-contact current measurement scheme and may be implemented with a small size, thus being available by being attached to an existing machine tool. Moreover, various sensor modules and communication modules may be attached, such that various types of data such as temperature, humidity, and vibration as well as 3-phase power may be synchronized and collected and a desired communication protocol (e.g., WiFi5, BLE3, ZigBee, etc.) may be selected for communication. In addition, light-weight artificial intelligence may be inserted for calculation at an edge and thus may be installed in small and medium manufacturing lines where a communication network is not properly established or a sufficient communication speed is not secured, thereby realizing intelligence of the manufacturing line without a burden of the amount of traffic. Furthermore, the modular sensor platform apparatus may be installed on a power distribution board in general home or inside a building, and a temperature sensor, a humidity sensor, an illuminance sensor, etc., may be attached thereto for establishment of intelligence, thus being applicable to a solution of a smart home or a smart building.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. **1** shows an example of an overall shape of a modular sensor platform apparatus according to an embodiment of the present disclosure.

(2) FIG. **2** shows an example of a configuration of a modular sensor platform apparatus according to an embodiment of the present disclosure.

(3) FIG. **3** shows an example of a calculation method used by a micro-control unit according to an embodiment of the present disclosure.

MODE FOR INVENTION

(4) Hereinafter, a modular sensor platform apparatus according to an embodiment of the present disclosure will be described in detail with reference to the accompanying drawings.

(5) FIG. **1** shows an example of an overall shape of a modular sensor platform apparatus according to an embodiment of the present disclosure.

(6) Referring to FIG. **1**, a modular sensor platform apparatus **100** may include at least one first terminal **110** that may be coupled to a power sensor and at least one second terminal **120** that may be coupled to an external sensor module and a communication module. The modular sensor platform apparatus **100** may include a micro-control unit that executes an algorithm predefined in a main body.

(7) As the modular sensor platform apparatus **100** includes the first and second terminals **110** and **120** that are connectable to various sensor modules and communication modules rather than including the sensor modules and the communication modules therein, the modular sensor platform apparatus **100** may be manufactured to have a small size and a light weight. Moreover, various types of sensor modules and various types of communication modules are connectable to the modular sensor platform apparatus **100**, thus being flexibly applicable to various environments. That is, each time when a physical quantity to be measured changes, the sensor module or communication module connected to the modular sensor platform apparatus **100** needs to be changed without replacing the modular sensor platform apparatus **100** or replacing or reassembling internal components thereof, thereby easily implementing the intelligence of conventional factories.

(8) The first terminal **110** may be connected to a non-contact current sensor that measures 3-phase power. Moreover, according to an embodiment, the first terminal **110** may be connected to various types of existing power sensors that measure power.
(9) The second terminal **120** may include a universal serial bus (USB) type and/or a 3-pole terminal. For example, the second terminal **120** may include at least one USB A type, at least one USB C type, and at least one 3.5 mm 3-pole terminal. For example, the communication module may be connected to a USB port, and a sensor that measures temperature, humidity, illuminance, vibration, etc., may be connected to a 3-pole terminal. The 3-pole terminal may receive an output signal from a connected resistance-based sensor and thus serve for data acquisition (DAQ) by reading the output signal without an additional micro-controller.
(10) According to another embodiment, the modular sensor platform apparatus **100** may include at least one image sensor, i.e., a camera, or may be connected to at least one camera through the second terminal **120**. In another example, the modular sensor platform apparatus **100** may be wirelessly connected to at least one camera through a communication module connected to the second terminal **120**.
(11) The modular sensor platform apparatus **100** may synchronize and collect sensing values of a plurality of sensor modules through the second terminal **120**. For example, the modular sensor platform apparatus **100** may collect the sensing values from the plurality of sensor modules at the same time.
(12) FIG. **2** shows an example of a configuration of a modular sensor platform apparatus according to an embodiment of the present disclosure.
(13) Referring to FIGS. **1** and **2** together, the modular sensor platform apparatus **100** may include a power metering integrated circuit (IC) **200**, a micro-control unit (MCU) **210**, and a universal asynchronous receiver/transmitter (UART) **220**.
(14) The power metering IC **200** may identify power by using a sensing value received from a power sensor through the first terminal **110**. For example, the power sensor, which is a current sensor of a non-contact type, may measure current flowing through each wire by being clamped to a 380 V 3-phase 4-wire power supply line. The power metering IC **200** may identify power, etc., based on a current sensing value of each wire, measured by the current sensor.
(15) The MCU **210** may be connected to at least one sensor modules **250** and **252** and at least one communication modules **260** and **262** through the second terminal **120**. In an embodiment, the MCU **210** may transmit sensing values received from the at least one sensor modules **250** and **252** connected in a wired communication manner through the communication modules **260** and **262** in a wireless communication manner, but in this case, the amount of traffic may increase. Thus, in the current embodiment, the MCU **210** may transmit a result of performing a calculation process based on the sensing values received from the at least one sensor modules **250** and **252** through the communication modules **260** and **262**. An example of calculation performed by the MCU **210** will be described again with reference to FIG. **3**.
(16) The MCU **210** may continuously receive a sensing value from a sensor module or determine to operate a sensor module. In an example, the MCU **210** may receive a control command from an external source through the communication modules **260** and **262** to determine the type and number of sensor modules to operate. That is, a user may transmit a control command for operating or stopping operating sensor modules connected to the modular sensor platform apparatus **100** through a server, a user terminal, etc. For example, when first through Nth sensor modules are connected through the second terminal **120**, the MCU **210** may operate first and fifth sensor modules upon receiving an operation command for the first and fifth sensor modules.
(17) In another example, the MCU **210** may determine whether to operate a sensor module according to a predefined condition. For example, when any one of the sensor modules is an image sensor, i.e., a camera, a lot of loads may be applied to a storage space or calculation in order for the modular sensor platform apparatus **100** to continuously process an image captured by the camera.

In this case, the MCU **120** may normally stop operating the camera, and when a power sensing value connected to the first terminal **110** is abnormal or a value measured from various sensor modules connected to the second terminal **120** corresponds to a predefined condition (e.g., an abnormal state, etc.) then the MCU **120** may operate a camera sensor module. That is, when an abnormal state is discovered in equipment or a space where the modular sensor platform apparatus **100** is installed, the image captured through the camera in this case may be transmitted to the server, the user terminal, etc.

(18) The communication modules **250** and **252** connected to or used for the modular sensor platform apparatus **100** may differ with a communication scheme of a place where the modular sensor platform apparatus **100** is located. For example, when ZigBee communication is required, a ZigBee communication module may be connected to the modular sensor platform apparatus **100**.

(19) In another embodiment, a plurality of communication modules **260** and **262** may be previously connected to the modular sensor platform apparatus **100** and when necessary, the MCU **210** may transmit and receive data through any one of the plurality of communication modules **260** and **262**. For example, for an environment where a plurality of communication protocols are available, the modular sensor platform apparatus **100** may select any one of the communication protocols according to user settings or a communication environment (e.g., a bandwidth, a communication delay, the amount of traffic, etc.), and transmit and receive data through a communication module supporting the selected communication protocol.

(20) The power metering IC **200** and the MCU **210** may be connected through the UART **220**. As the power metering IC **200** and the MCU **210** are physically separated from each other and transmit and receive data through the UART **220**, the MCU **210** may be protected in spite of sudden occurrence of an overload, etc., in the power metering IC **200**.

(21) FIG. **3** shows an example of a calculation method used by a micro-control unit according to an embodiment of the present disclosure.

(22) Referring to FIG. **3**, the MCU **210** may include an algorithm **300** that may be calculated based on a sensing value, thus serving for a sort of edge computing. The algorithm **300** may be stored in an internal memory of the MCU **210** or a separate memory present in the modular sensor platform apparatus **100**.

(23) The MCU **210** may derive an analysis result **320** by inputting a power measurement value **310** identified through a power sensor and sensing values **312** and **314** measured through respective sensor modules, etc., to the algorithm **300**. The algorithm **300** may be implemented as a general function, etc., or an artificial intelligence algorithm based on deep learning or machine learning. For example, after attached to a machine tool, etc., the modular sensor platform apparatus **100** may identify whether the machine tool is abnormal by using the power measurement value **310** and the sensing values **312** and **314**, and transmit the analysis result **320**, thereby reducing the amount of traffic and a calculation load of a central server when compared to transmitting the sensing values **312** and **314**.

(24) In another embodiment, operating conditions of various sensor modules may be predefined in the algorithm **300**. For example, the MCU **210** may normally operate some of the plurality of sensor modules according to the algorithm **300** to identify whether the machine tool, etc., is abnormal, and may operate the camera upon detection of abnormality through the algorithm **300** to transmit the captured image to the server, the user terminal, etc.

(25) According to another embodiment, the MCU **210** may receive a control command of whether to operate a specific sensor module from the user through a communication module. For example, the MCU **210** may receive the control command of whether to operate the camera from an external source, operate the camera according to the control command, and transmit the captured image.

(26) So far, embodiments have been described for the disclosure. It would be understood by those of ordinary skill in the art that the disclosure may be implemented in a modified form within a scope without departing from the essential characteristics of the disclosure. Therefore, the disclosed

embodiments should be considered in a descriptive sense rather than a restrictive sense. The scope of the present specification is not described above, but in the claims, and all the differences in a range equivalent thereto should be interpreted as being included in the disclosure.

Claims

1. A modular sensor platform apparatus comprising: at least one first terminal capable of being coupled to a power sensor; a plurality of second terminals capable of being coupled to an external sensor module and a communication module; and a micro-control unit (MCU) configured to execute a predefined algorithm, wherein the MCU is further configured to operate a camera connected to one of the plurality of second terminals and to transmit an image captured through the camera when a power sensing value of the power sensor received through the first terminal is abnormal.
2. The modular sensor platform apparatus of claim 1, wherein the first terminal is connected to a non-contact current sensor configured to measure 3-phase power.
3. The modular sensor platform apparatus of claim 1, wherein the second terminal comprises a universal serial bus (USB) type terminal.
4. The modular sensor platform apparatus of claim 1, wherein the second terminal comprises a 3-pole terminal, and the 3-pole terminal is configured to receive an output signal from a connected resistance-based sensor module.
5. The modular sensor platform apparatus of claim 1, wherein the algorithm is an artificial intelligence algorithm based on deep learning, and the MCU is configured to transmit, through a communication module, a result obtained by inputting sensor values received through the first terminal and the second terminal to the algorithm.
6. The modular sensor platform apparatus of claim 1, further comprising a power metering integrated circuit (IC) configured to meter power based on a signal received through the first terminal, wherein the power metering IC and the MCU are connected to each other through a universal asynchronous receiver/transmitter (UART).